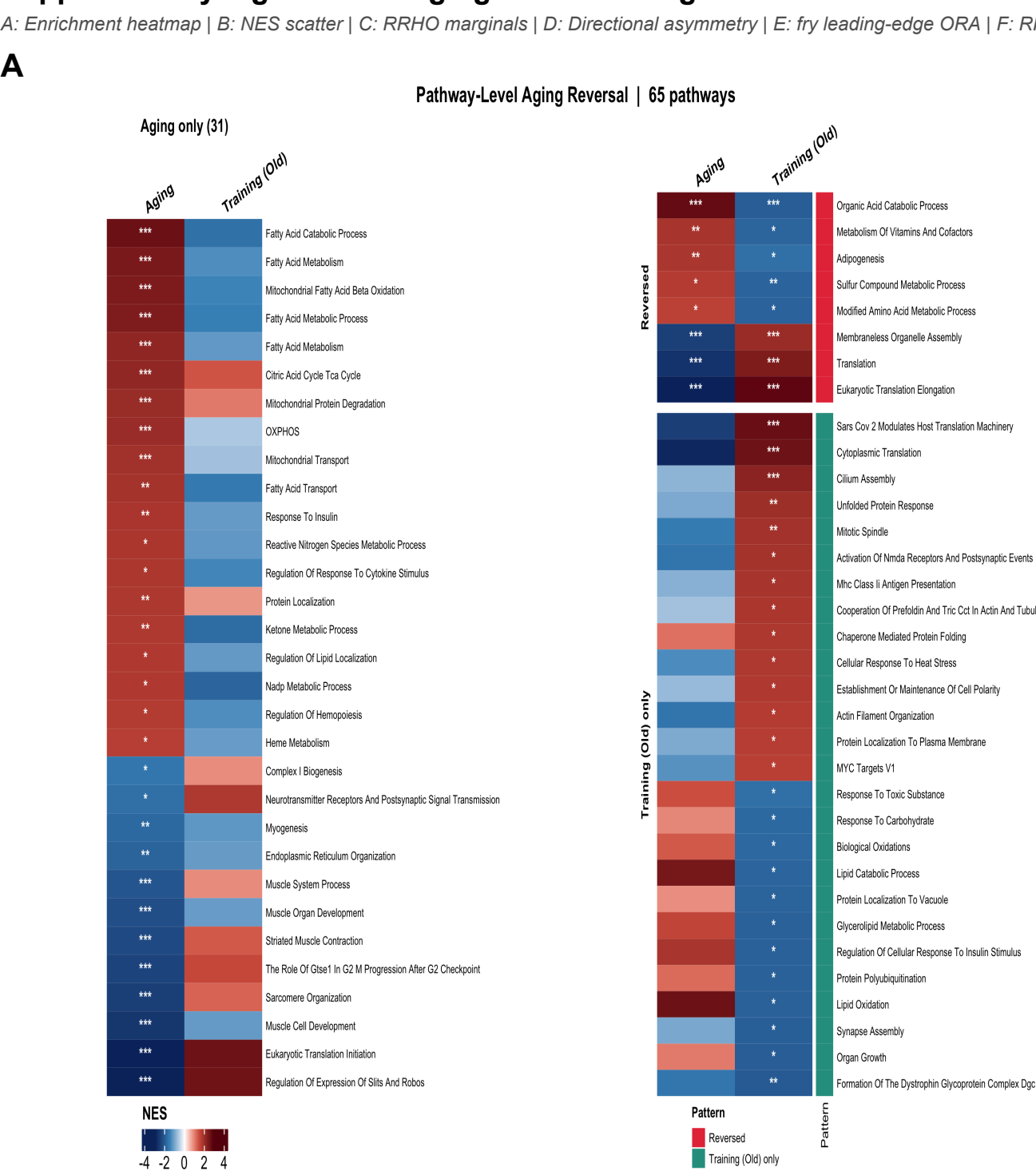


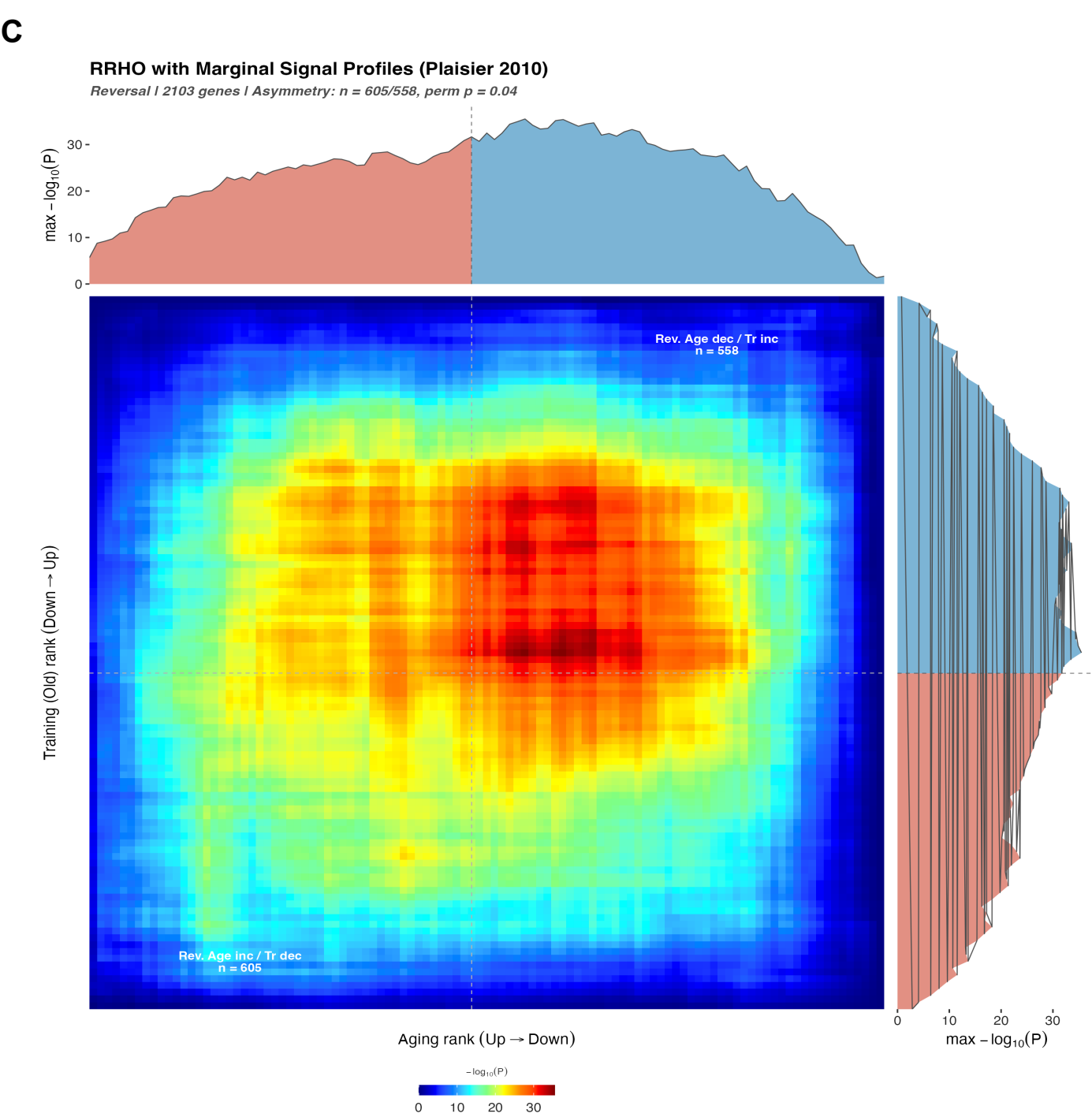
A: Enrichment heatmap | B: NES scatter | C: RRHO marginals | D: Directional asymmetry | E: fry leading-edge ORA | F: RRHO2 ORA

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A:



C

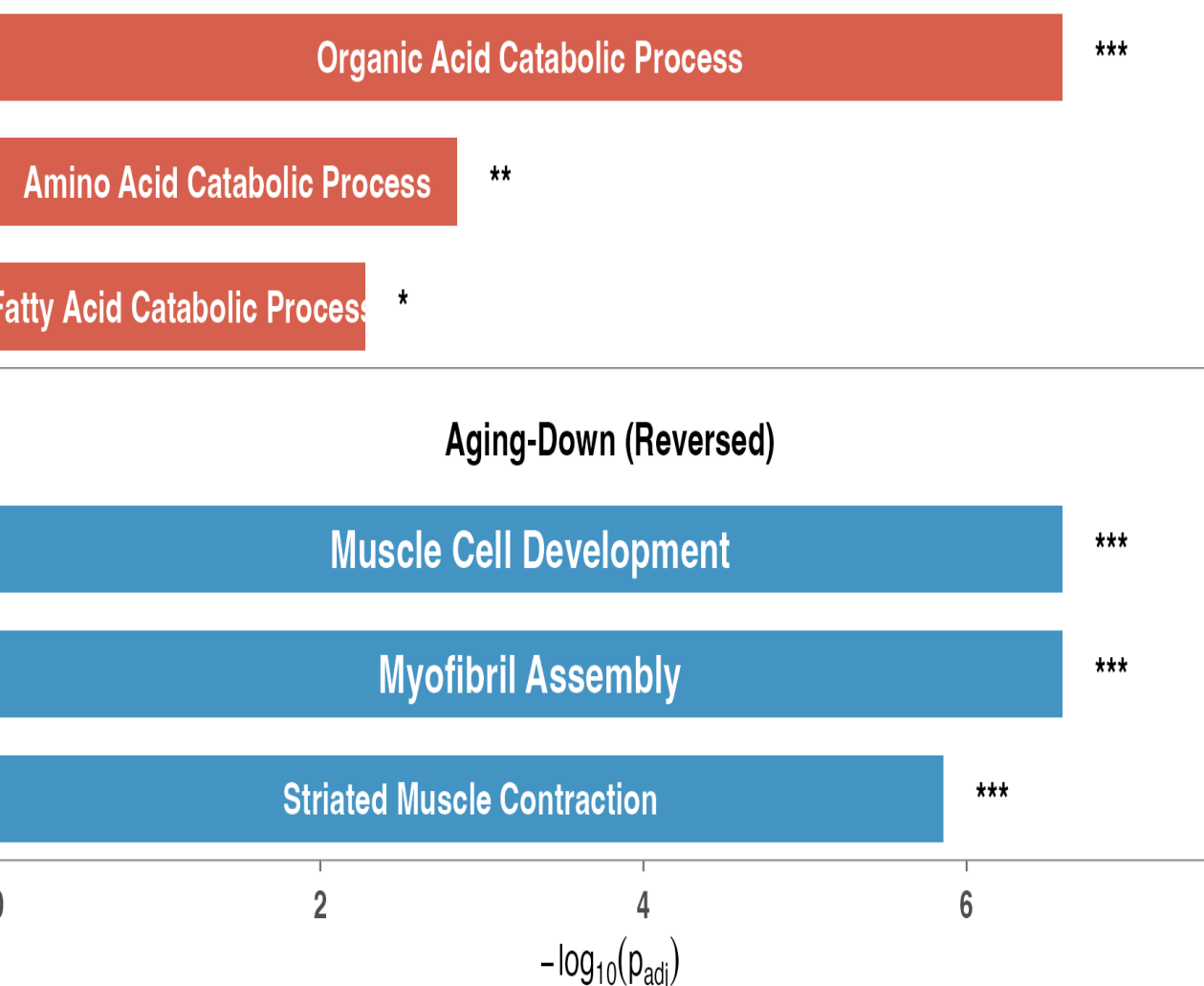


E

Leading-Edge ORA: fry Driving Proteins (Reversal)

Hypergeometric ORA on reversal-driving proteins | top 3 per set

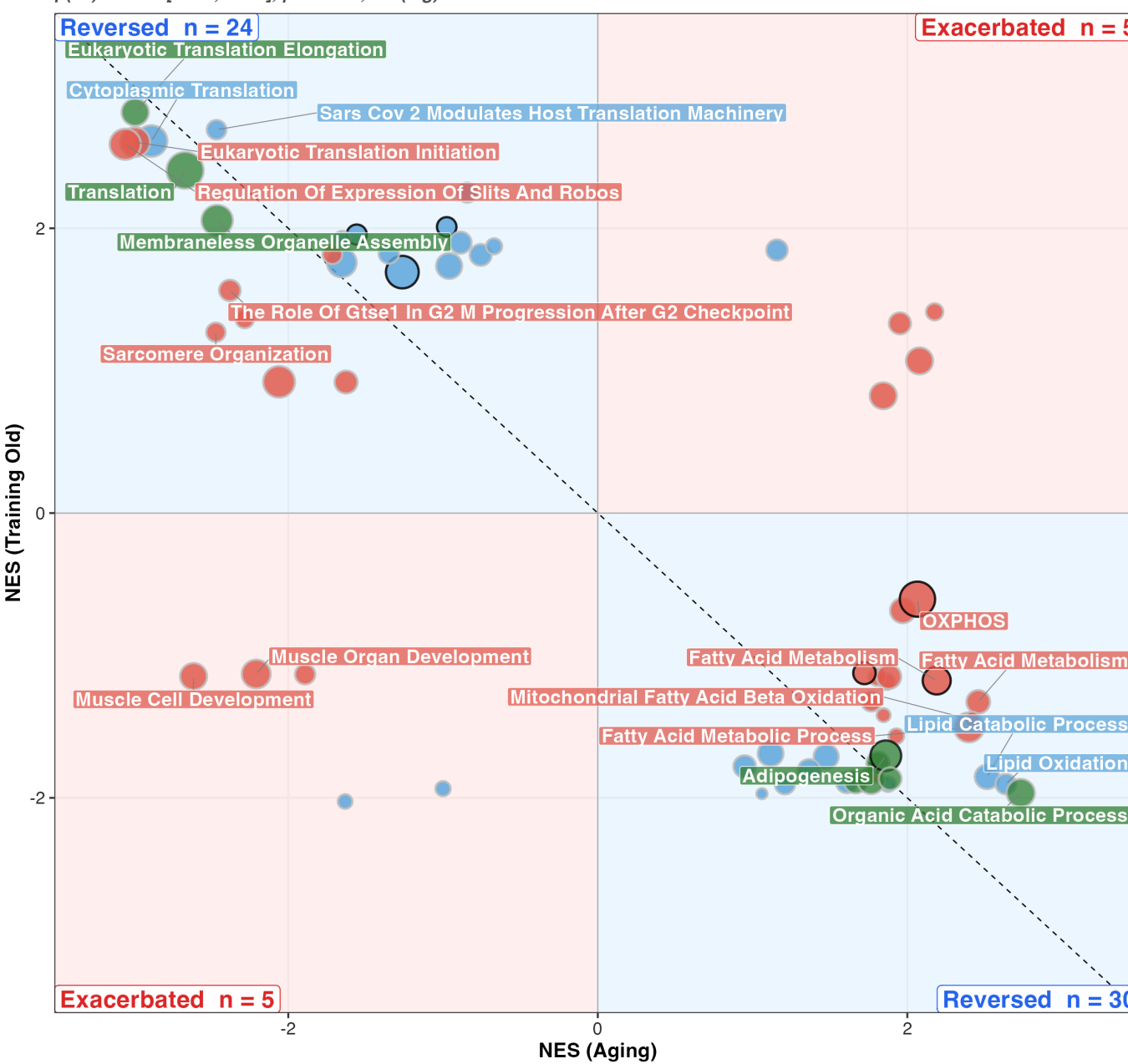
Aging-Up (Reversed)



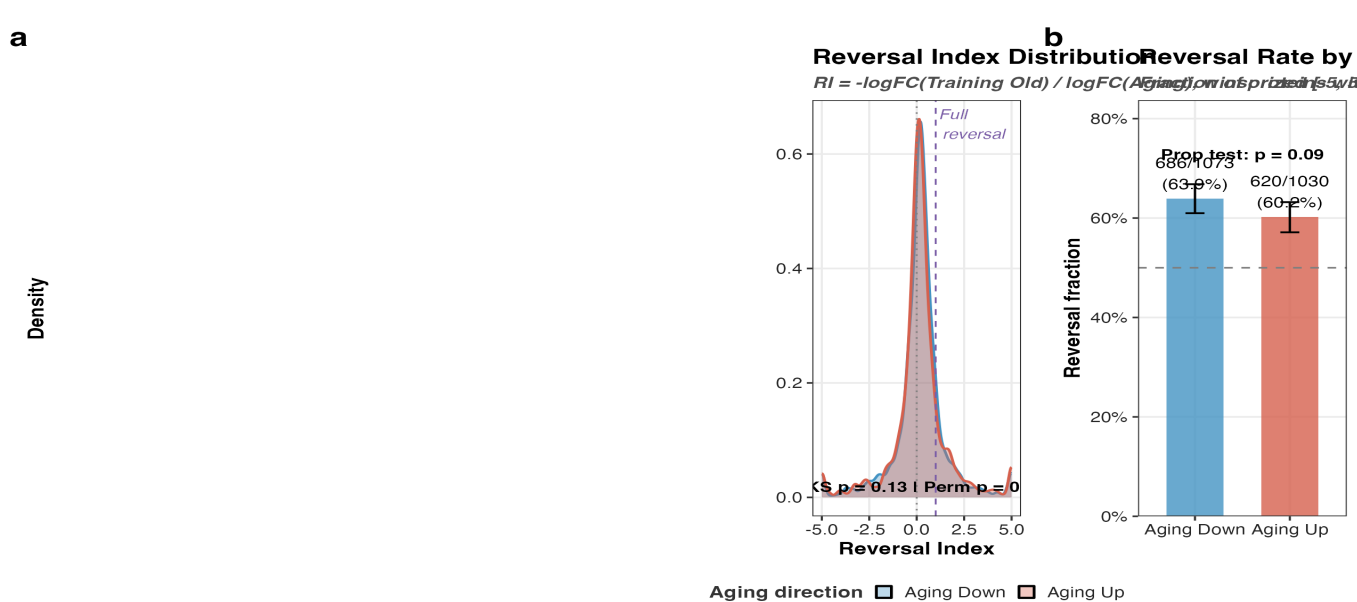
B

Pathway-Level Reversal (fGSEA, collapsePathways)

H + KEGG + Reactome + GO:BP (collapsePathways) | 64 pathways (64 sig.) | fGSEA on limma t-statistics
 $\rho(\text{all}) = -0.59 [-0.73, -0.40]$, $p < 0.001$, $\rho(\text{sig}) = -0.59$ | 84% reversed



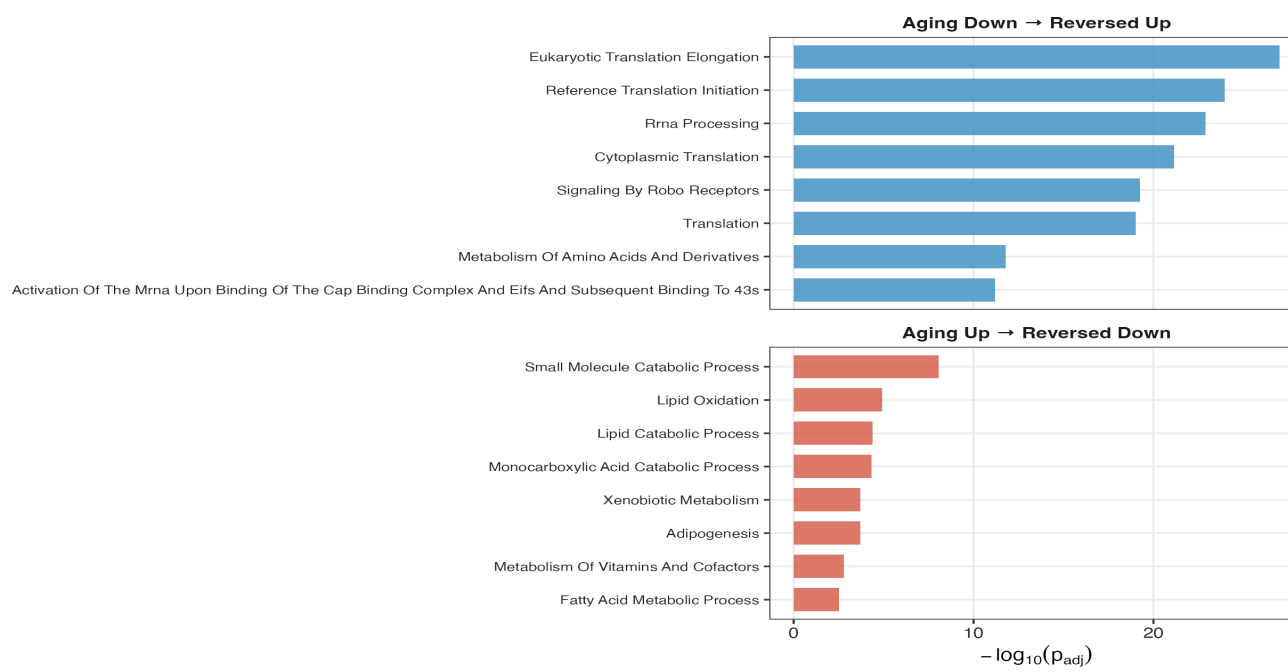
D



C

Direction-Stratified Pathway Enrichment

Top 8 ORA pathways per reversal quadrant (padj < 0.05)



F

Threshold-Free Reversal (RRHO2)

Stratified hypergeometric / 2103 shared genes / warm off-diagonal = training reverses aging / No MTC (Cahill et al. 2018)

